

Aug 18 Review I.



ideals in commutative rings $I \subseteq R$.

1. nonempty subsets that contains 0

and it is closed under addition and

multiplication by a ring element.

$$\bullet \quad I = R \quad \text{iff} \quad 1_R \in I$$

$$\text{iff} \quad I = (u) \quad \text{for } u \text{ an unit in } R.$$

2. ideals characterizes rings.

R is a field iff R has only

two ideals. (0) and R itself.

R is PID if every ideal is

generated by one element.

3. more specific ideals

I is prime ideal if $ab \in I \Rightarrow a \in I$ or $b \in I$

I is maximal ideal if $I \subseteq J \Rightarrow J = I$ or $J = R$

R is a field iff (0) is a maximal ideal.

R is a domain iff (0) is a prime ideal.

4. ideals characterize elements

(p) is a prime ideal iff p is prime in R

(π) is maximal among principal ideals

iff π is irreducible in R

5. rings determines ideals

$\mathbb{F}[D]$ is PID.

In PID, nonzero prime ideals are maximal

• elements determines ideals

factorization gives chain of ideals

6. induced structures

integer congruences: $\mathbb{Z}_m = \mathbb{Z} \bmod m$.

polynomial congruences: $\mathbb{R}[x] \bmod x^2+1$

irreducibility in $\mathbb{Q}[x]$ and $\mathbb{Z}[x]$.

.. irreducibles in $\mathbb{Q}[x]$.

$\mathbb{Q}[x]$ is an ED $\Rightarrow$ PID $\Rightarrow$ UFD.

so irreducibles are the primes, i.e.

$f(x)$ of positive degree is irreducible iff $(f(x))$ is prime ideal

iff $(f(x))$ is maximal ideal.

iff $f(x)$ can't be factored into a product

$f(x) = g(x)h(x)$ where $g(x), h(x) \in \mathbb{Q}[x]$,

and $\deg g(x), \deg h(x) < \deg f(x)$

$f(x)$ is of deg 0, iff $f(x)$ is a unit.

2. irreducibility in $\mathbb{Z}[x]$.

irreducibles in $\mathbb{Z}[x]$ are precisely

(1) $\pm p$ for prime $p \in \mathbb{Z}$.

(2) $f(x)$ of positive degree that is
primitive and irreducible in $\mathbb{Q}[x]$.

(2') $f(x)$ of positive degree is irreducible in $\mathbb{Z}[x]$

iff $f(x)$ is primitive and $f(x)$ irreducible in $\mathbb{Q}[x]$

(2'') $f(x)$ of positive degree. primitive

is irreducible in $\mathbb{Z}[x]$ iff is irreducible in $\mathbb{Q}[x]$

iff it can't be factored into a product of
polynomials with strictly smaller degrees

3. factorize $f(x)$ in $\mathbb{Q}[x]$

$p \in \mathbb{Q}$ is a root of $f(x)$

iff $(x-p) \mid f(x)$

rational root test: $\frac{p}{q}$ is a root

to $a_n x^n + \dots + a_0$ for $a_n, \dots, a_0 \in \mathbb{Z}$

if $p \mid a_n$ and $q \mid a_0$

irreducibility test.

reduction mod p

Eisenstein criterion.

4. Factorize $f(x)$ in $\mathbb{Z}[x]$.

first factor out the gcd. of coefficients

and put $f(x) = m \cdot p(x)$ for

$m \in \mathbb{Z}_+$ and $p(x)$ primitive.

Then. factorize $p(x)$ in $\mathbb{Q}[x]$

because most likely $\deg p(x) > 0$ and

we know $p(x)$ irreducible in $\mathbb{Q}[x]$

iff $p(x)$ irreducible in $\mathbb{Z}[x]$.

5. Why irreducibles in $\mathbb{Z}[x]$ is stronger?

irreducibles corresponds to principle ideals

that is maximal among all principle ideals

$\mathbb{Q}[x]$ is PID. whereas $\mathbb{Z}[x]$ is not.

Also $\mathbb{Z}[x]$ is UFD. so prime ideals

corresponds to irreducibles. Since $\mathbb{Z}$

is a domain not a field. prime ideals

have more forms. eg. (2) . $(2, x)$.

